Mechanical–Hydraulic Modeling of an Embankment Mower in Simscape

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***Abstract*—Hydraulic manipulators couple configuration-dependent mechanical loads with compressible-fluid dynamics and finite valve response. This project develops a Simscape model of the MULAG FME 500 embankment mower by combining CAD-derived geometry, a multibody mechanism, and valve-controlled hydraulic actuation. Cylinder geometry maps piston motion and axial force to joint motion and generalized torque. A joint-level feedback controller provides a repeatable trajectory-tracking task for studying plant sensitivity. Reported simulation results at four bulk moduli, from 0.5 to 2.0 GPa, show a decrease in the dominant joint's 50% response delay from 847.4 to 649.7 ms and a decrease in end-effector RMS tracking error from 827.9 to 739.4 mm. The effects are not uniform: the fourth monitored joint's delay increases, while the mean tracking error is higher at the largest modulus than at the smallest. Numerical trajectory curves and tabulated results are preserved and redrawn as vector figures. The report details cylinder assembly, liquid and mechanical ports, valve routing, the shared hydraulic network, and the reference/measurement paths. A block-level specification distinguishes active components from disabled alternatives and identifies parameter and unit checks needed for reproducibility. The available results establish reported simulation sensitivity rather than quantitative agreement with measured hardware.**

***Index Terms*—Hydraulic manipulator, multibody dynamics, Simscape, bulk modulus, trajectory tracking, virtual prototyping.**

I. Introduction

An embankment mower must position a cutting head through a long articulated mechanism while hydraulic actuators support gravitational, inertial, and external loads. An ideal joint-position or torque source is useful for kinematic analysis, but it does not represent pressure buildup, actuator compliance, flow limits, or valve dynamics. These effects matter when studying the time required to initiate motion and the error between a requested and realized cutting-head path.

Hydraulic control models commonly combine fluid continuity, pressure-dependent orifice flow, and actuator mechanics [1], [2]. The modeling challenge is to connect these relations to the actual linkage geometry. A cylinder's moment arm changes with configuration, and motion of the linkage changes the fluid volume in each chamber. The resulting interaction is bidirectional: hydraulic force drives the mechanism, while mechanical motion changes chamber flow and pressure.

This project develops a coupled mechanical–hydraulic model of the MULAG FME 500 using MATLAB, Simulink, Simscape Multibody, and Simscape Fluids. The mechanical structure is derived from CAD and URDF assets; hydraulic actuators replace idealized actuation at the physical interface. A fixed joint-level controller is used to investigate the sensitivity of the simulated plant to bulk modulus. The contribution is the integrated modeling workflow and its documented sensitivity study, rather than a new controller stability theorem.

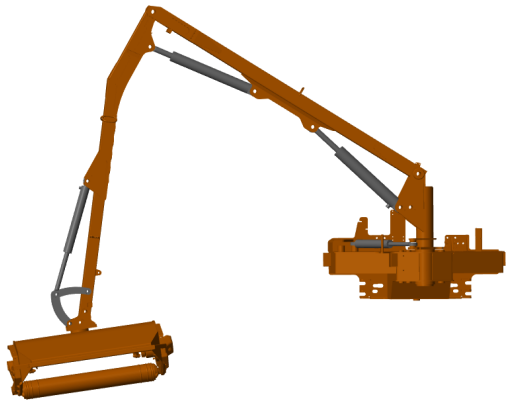


Fig. 1. CAD-derived visualization of the MULAG FME 500 mechanism supplied with the project. The independent arm coordinates are distinguished from the additional bodies and joints used to represent cylinder assemblies.

The report develops the assembly procedure, network construction, governing equations, and controller implementation, then evaluates the saved numerical curves and reported sensitivity cases. It introduces no new simulation or hardware results.

II. Mechanical Model

A. Geometry and Model Import

The reference kinematic description uses five joint coordinates $q \in \mathbb{R}^5$. The project supplies several URDF descriptions: a six-link, five-revolute-joint representation for arm kinematics and a more detailed tree including actuator bodies. Extra joints in the detailed URDF represent actuator bodies and alignments; they should not be interpreted as additional independently controlled arm degrees of freedom. The saved dynamic model also constrains the final cutting-head joint to a

very narrow interval, so its nominal five-coordinate description does not imply five unrestricted axes in every configuration.

The CAD-to-URDF workflow provides link geometry, attachment frames, masses, and inertia tensors. URDF and rigid-body-tree models can be imported into Simscape Multibody using the supported model-import tools [3]. Inertial cross terms and the reference frame of each tensor must be preserved consistently. Geometry alone does not establish a measured mass distribution: density assignments, inertia reference frames, and imported units remain calibration inputs.

Figure 2 identifies the imported links, revolute coordinates, and cylinder installations. The base-swing motion is driven by cylinder 1, the first and second booms by cylinders 2 and 3, and the cutting-head swivel linkage by cylinder 4. The final cutting-head coordinate has its own command path. The saved arm-joint bounds are $[-1.57, 0.52]$ rad for the base swing, $[-0.52, 0.52]$ rad for the two boom joints and swivel joint, and $[0, 10^{-6}]$ rad for the final head joint. These are active model limits, not manufacturer-certified travel ranges. In particular, the last interval is effectively locked and differs from the controller's much wider reference-clipping interval.

For a chosen external-wrench convention, the joint-space equations may be written

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) + \tau_f(\dot{q}) = \tau_a + J_e(q)^{\mathsf{T}} w_e, \quad (1)$$

where M is the inertia matrix, $C\dot{q}$ collects velocity-dependent terms, g is gravity torque, and w_e is the wrench applied by the environment to the end effector. The sign of the external term reverses if the opposite wrench convention is used. The model must use a consistent frame and sign convention throughout the physical interfaces.

B. Cylinder Kinematics and Power Transfer

Let $a(q)$ and $b(q)$ be the cylinder attachment positions in a common frame. The cylinder length, extension, and axial direction are

$$\ell(q) = \|b(q) - a(q)\|_2, \quad (2)$$

$$s(q) = \ell(q) - \ell_0, \quad (3)$$

$$e_c(q) = \frac{b(q) - a(q)}{\ell(q)}. \quad (4)$$

If J_a and J_b map joint velocities to attachment-point velocities, then

$$J_c(q) = e_c(q)^{\mathsf{T}}[J_b(q) - J_a(q)], \qquad \dot{s} = J_c(q)\dot{q}. \quad (5)$$

Virtual work gives the generalized torque contribution

$$\tau_a = J_c(q)^{\mathsf{T}} F_h. \quad (6)$$

This map preserves instantaneous mechanical power: $\tau_a^{\mathsf{T}}\dot{q} = F_h\dot{s}$. For multiple cylinders, the scalar extension rows are stacked into a matrix. Geometry-dependent leverage is therefore included without assuming a constant force-to-torque ratio.

Cylinder attachments form closed mechanical loops in the assembled multibody model. The active actuator subsystems use spherical, prismatic, and cylindrical joint connections to provide axial extension and alignment freedom. Joint-frame inspection is essential because an incompatible connection can overconstrain a loop or change the sign of the actuator stroke. Section III details the actual construction and the separation between multibody frames, translational conserving ports, and physical signals.

C. Hydraulic Stiffness in Joint Coordinates

For a local closed-valve approximation with positive chamber volumes, the hydraulic contribution to axial stiffness is

$$k_h \simeq \beta_{\text{eff}} \left(\frac{A_A^2}{V_A} + \frac{A_B^2}{V_B} \right). \quad (7)$$

The leading material-stiffness contribution in joint coordinates is

$$K_{q,\text{mat}} \simeq J_c^{\mathsf{T}} k_h J_c. \quad (8)$$

The complete derivative of (6) can also contain geometric-stiffness terms from the configuration dependence of J_c. Thus this expression is a local approximation, not a complete global stiffness matrix. Leakage, open valves, structural flexibility, and operating pressure alter the effective response.

The combination of slow arm motion and faster pressure dynamics can produce a stiff differential-algebraic system. The archived integrated model stores the implicit variable-step solver ode15s; its stored relative tolerance is 10^{-3} and maximum step is 1.2 s. These are archived settings, not a verified accuracy recommendation. Step-size and tolerance convergence should be checked for each result [4].

III. Actuator Construction and Multibody Interface

A. Subsystem Boundary and Repeated Structure

Each of the four hydraulic axes is built as an Actuator_i subsystem. The boundary contains one Simulink command input, two multibody frame connections, and two isothermal-liquid connections named P and T. Inside this boundary are the directional valve, double-acting actuator, an external translational friction element, barrel and rod CAD bodies, alignment joints, the prismatic actuation joint, pressure taps, and force/extension sensing. This organization makes the hydraulic drive and its mechanical installation one reusable unit rather than an ideal torque applied to an arm joint.

Table I relates the actuator names to the imported mechanical parts. The fifth command follows a different implementation: Actuator_5 is a Simulink–PS conversion feeding the cutting-head revolute joint's torque input. It contains no pump connection, valve, or double-acting cylinder. Accordingly, the integrated model has five command channels but four hydraulic actuator branches.

B. Building the Mechanical Cylinder Assembly

The barrel (Rohr) and rod (Kolben) remain separate solid bodies so that their imported mass, inertia, and attachment frames participate in the multibody equations. Rigid transforms place joint frames on the CAD attachment axes; the

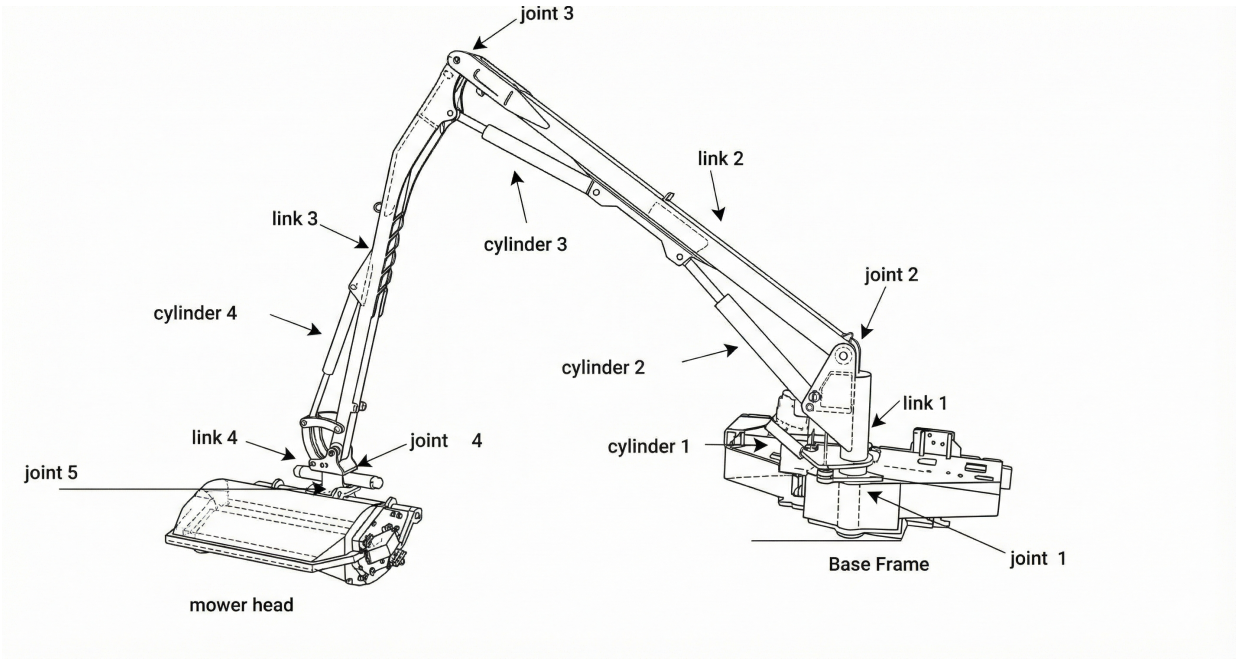


Fig. 2. Original annotated mechanical topology: base frame, five nominal revolute coordinates, arm links, and four cylinder installations. The image identifies physical placement; the selected model's active joint limits and direct fifth-axis command must additionally be considered when interpreting the available degrees of freedom.

TABLE I
ACTUATION BRANCHES AND MECHANICAL INSTALLATION IN THE SAVED MODEL

Command	Subsystem	Imported cylinder/mechanism	Mechanical role
u_1	Actuator_1	Ausleger_Schwenkzylinder	Base-to-swing installation
u_2	Actuator_2	Hebezylinder	Swing assembly to first boom
u_3	Actuator_3	Knickzylinder	First boom to second boom
u_4	Actuator_4	Messerkopf_Schwenkzylinder	Second boom to cutting-head swivel linkage
u_5	Actuator_5	Messerkopf_link_joint	Direct physical-signal torque command

visible mesh alone does not define the cylinder axis. The spherical connection attaches the barrel assembly to the upstream link and permits alignment. The internal prismatic joint permits rod translation along the cylinder axis. The rod-side cylindrical connection provides angular alignment while its axial coordinate is constrained by very tight stored limits. It is therefore not intended to introduce a second freely extending stroke.

The main arm retains its revolute joints. The cylinder forms a second mechanical path between two arm links, so changing cylinder extension changes the revolute configuration through loop closure. An imposed joint-angle trajectory and a force-driven cylinder must not independently prescribe incompatible motion to the same closed loop.

The construction sequence is to import the arm and attachment frames, insert the separate barrel/rod bodies, align the prismatic axis, connect the spherical and cylindrical ends, and check assembly in the initial configuration. Next, enable prismatic position/velocity sensing and force input, then connect the hydraulic interface. Finally, check the sign of a small commanded extension against the resulting joint motion. This sequence separates frame-assembly problems from hydraulic initialization and controller tuning.

C. Connecting Hydraulic Power to the Prismatic Joint

The Double-Acting Actuator (IL) has liquid ports A/B and translational conserving ports C/R. C represents the casing and R the rod. In the saved model, C is connected to the translational reference and to the interface casing terminal; R is shared by the actuator, external friction element, and interface rod terminal. This reference fixes the scalar translational coordinate system. It does not ground the moving barrel CAD body in the three-dimensional arm.

The Translational Multibody Interface receives the prismatic joint's sensed relative velocity at its physical-signal input v and supplies the force signal f to that joint's force-actuation port [5]. The joint's sensed position is also connected directly to the actuator's p input and to the extension sensor. The actuator is configured to receive displacement from the multibody joint, rather than independently integrating a second position state [6]. Thus the position-dependent chamber volume uses the same motion that produces translational velocity.

For an ideal interface the transmitted quantities satisfy

$$P_{\mathrm{trans}} = F_{\mathrm{int}} v_c, \qquad v_c = \dot{s}_{\mathrm{joint}}. \tag{9}$$

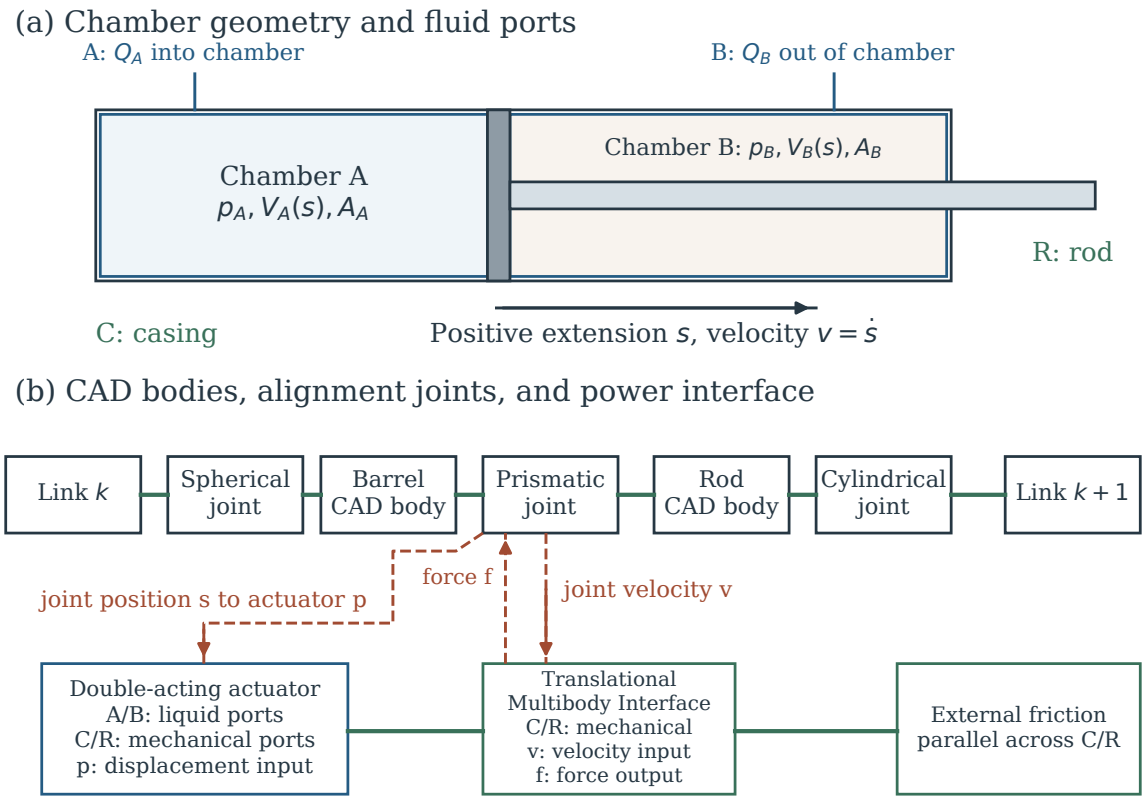


Fig. 3. Actuator construction. (a) Symbolic chamber geometry, fluid ports A/B, casing C, rod R, and positive extension. Chamber areas are symbolic; the saved active blocks use equal A/B areas as listed in Table II. (b) CAD barrel and rod bodies form a closed chain between adjacent arm links through spherical, prismatic, and cylindrical joints. The prismatic joint supplies position and velocity; the translational interface returns axial force. Solid green connections represent mechanical coupling and dashed red connections represent physical signals. The friction element acts in parallel across the actuator's translational terminals.

The frame ordering, mechanical orientation, and signal signs must be consistent with this relation. Reversing a sensed velocity without reversing the corresponding force convention changes the apparent power transfer. The force log denotes the interface signal, not an independently measured external load.

D. Cylinder Geometry, Initial State, and Stroke Limits

Positive extension increases chamber A volume and decreases chamber B volume. Dead volumes keep the pressure states finite near a stroke end. For a conventional single-rod cylinder, geometric areas can be obtained from bore diameter d_b and rod diameter d_r as

$$A_A = \frac{\pi d_b^2}{4}, \qquad A_B = \frac{\pi (d_b^2 - d_r^2)}{4}. \tag{10}$$

However, these geometric relations must not be used to silently replace the actual saved block values. The active cylinders in this model use equal A/B effective areas. The general asymmetric-cylinder equations remain useful, but the stored configuration is an equal-area parameterization rather than evidence of an identified single-rod bore.

Table II records values and units read directly from the active blocks; the pressure entries are stored initialization parameters and are not measured operating pressures or guaranteed enforced initial conditions. The piston offset is the expression (hydr_x0)-hydr_stroke. Its evaluated value and initialization priority must be checked with the model workspace. The external friction blocks have breakaway/Coulomb forces of 200,000/200,000 N on branches 1/2, 1,000/1,000 N on branch 3, and 2,000/2,000 N on branch 4; the stored breakaway speed is 0.060623 m/s. These large effective areas and friction settings should be reconciled with geometry and units before a new physical-accuracy claim is made.

Dynamic chamber compressibility is enabled in all four active actuator blocks. Their built-in cylinder-friction, internal-leakage, and end-cushion options are disabled; friction is instead represented by the separate parallel element. The prismatic joints on branches 2–4 also store 5,000 N s/m damping, while branch 1 stores zero. This damping is an additional mechanical contribution and must not be confused with the external friction coefficient.

The prismatic lower/upper limit flags are enabled only on branch 2, at -0.2 and 0.2 m. Branches 1, 3, and 4 retain numerical limit fields with their activation flags off. The actuator's own stroke/hard-stop settings and the cylindrical alignment-joint limits remain separate constraints. Stored bound fields alone do not activate a joint limit.

TABLE II
LITERAL ACTIVE CYLINDER AND EXTERNAL FRICTION SETTINGS

Branch	$A_A = A_B$ (m^2)	Stroke (m)	$V_{A0} = V_{B0}$ (m^3)	Stored p_{A0} (MPa)	Stored p_{B0} (MPa)	b_v (N s/m)
1	0.2513274	1.85	10^{-4}	3.548700	24.232950	10^8
2	0.2513274	1.85	10^{-4}	3.548700	24.232950	10^8
3	0.1256600	2.25	10^{-4}	7.685550	13.890825	10^7
4	0.1256600	2.25	10^{-4}	7.685550	13.890825	10^7

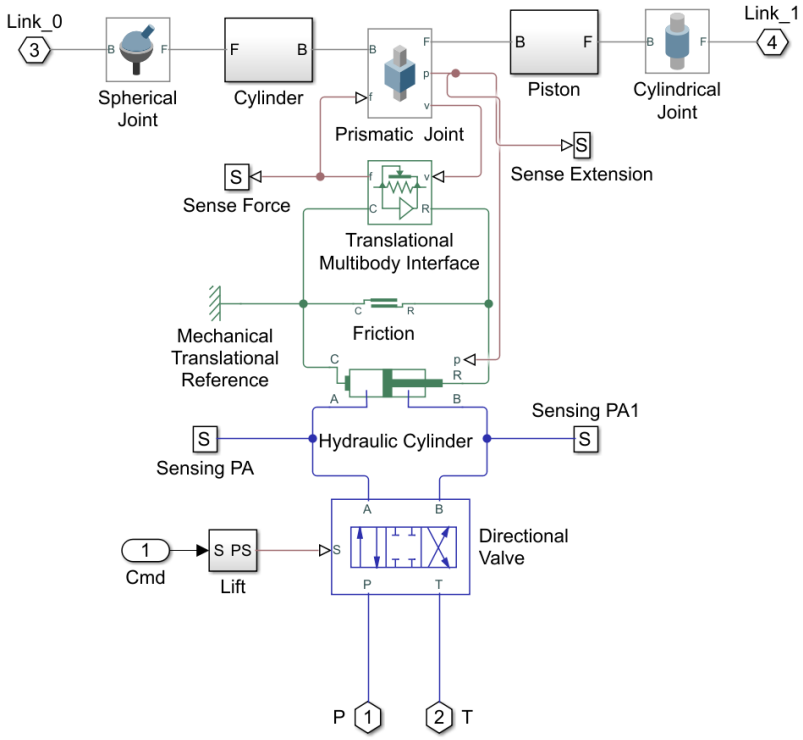


Fig. 4. Original project view of a complete electro-hydraulic actuator subsystem. Command conversion drives valve S; P/T join the common rails, and A/B join the sensed cylinder chambers. The middle network contains the translational actuator, parallel friction, reference, and interface. The upper closed chain contains the barrel, force-actuated prismatic joint, rod, and alignment joints. The physical-signal connections return position/velocity and apply force. The image records the project construction rather than a new run.

IV. HYDRAULIC NETWORK AND SIGNAL INTEGRATION

A. Complete Circuit Organization

The documented circuit is a shared supply/return arrangement with a fixed-displacement pump, reservoir, pressure-relief path, fluid-properties definition, pressure/flow sensing, and four parallel actuator modules. The pump converts imposed shaft rotation to fluid flow. Each valve meters this common flow into one cylinder chamber while allowing fluid from the other chamber to return. A relief path limits pressure only when its orientation, setpoint, and connection to the pressure rail are correct. The accumulator is part of the documented design, but its block is commented out in the saved editable model.

A shared manifold couples the hydraulic axes even when the controller is decentralized. Neglecting density variation for a local volume balance, a supply-node representation is

$$C_s \dot{p}_s = Q_p - \sum_{i=1}^{4} Q_{P,i} - Q_{\mathrm{rel}} - Q_{\mathrm{acc}}, \tag{11}$$

where C_s is an effective hydraulic capacitance and Q_{acc} is positive when liquid enters an enabled accumulator. This is an explanatory lumped balance, not a claim that a separate capacitance block exists on the saved rail. It shows why opening one valve can affect the pressure and available flow at another. Setting the accumulator block inactive removes its dynamic storage contribution from that configuration.

B. Saved Connections and Component Activation

Figure 6 reconstructs the saved top-level connection groups without assuming that the presentation image is the latest wiring. N1 denotes the rail connected to the atmospheric reservoir and each actuator's serialized RConn2 endpoint. N2

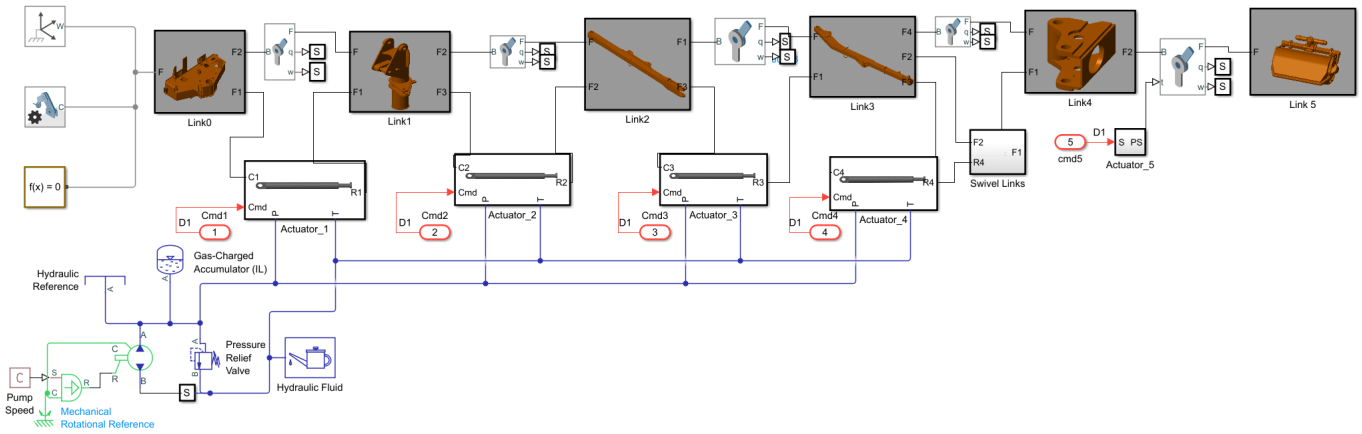


Fig. 5. Original integrated-model overview from the project presentation. The upper CAD chain is driven by four cylinder subsystems sharing two liquid rails; a fifth command drives the final revolute joint. The lower-left network includes pump drive, reservoir, relief, fluid properties, sensing, and an accumulator in the documented design. Figure 6 and Tables III–IV record the saved configuration and its disabled alternatives.

denotes the rail connected to the pressure/flow sensor, active fluid-properties block, and each actuator's RConn3 endpoint. The pump's serialized liquid-port connections join these two rails; the active relief block also connects across them. This endpoint notation deliberately preserves the static record so that physical supply/return direction can be checked in MATLAB rather than inferred from icon placement.

The Connection Port blocks inside each actuator are named Ci, Ri, P, and T, with stored port numbers 1–4. Because left/right connector indices, flexible port placement, and rotated library blocks are distinct representations, the association of the named P/T ports to each top-level rail must be confirmed with the loaded model [7]. The saved rail groups and the original presentation are not sufficient to certify pump rotation, relief opening direction, or pressure-boundary placement. A reproducible run should record compiled port handles and a small open-loop pressure/flow test before closing the position loop.

C. Supply Parameters and Unit Consistency

The presentation gives a nominal flow scale of 40 L/min and a pressure range of 280–360 bar. These are design-context values, not a complete parameterization of the saved pump. Table IV lists the active record. In particular, the pump displacement field is stored as 25 with unit m^3/rad. Combining that literal field with an interpreted 188 rad/s speed would produce an implausible geometric-flow scale; it must not be relabeled as 25 cm^3/rev without checking and correcting the model. The report records this discrepancy rather than claiming the stored pump realizes 40 L/min.

The active fluid-properties block stores a constant liquid bulk-modulus option and the air-related parameters. A stored field is not proof that every option-dependent effect is active. A new bulk-modulus sweep must identify the particular active block field being varied, preserve all other settings, and save the evaluated parameter set. Likewise, the nominal 34 MPa supply value in a helper file is not the same quantity as the active relief differential setpoint of 27.579 MPa.

D. Valve Routing and Command Conversion

Figure 7 expands the valve symbol. The active directional-valve configuration lists P–A and B–T for positive spool displacement, all nominal paths closed in neutral, and P–B and A–T for negative displacement [8]. These are valve-relative paths; extension direction still depends on the cylinder orientation, port connection, and linkage geometry. A nominally closed center also retains the finite stored leakage opening and does not establish exact load holding.

The four active valves share a maximum orifice-area field of 500 mm^2, leakage-area field of 10^{-12} m^2, discharge coefficient 0.7, and critical Reynolds number 12. The common spool-travel expression is approximately 0.005 m; several path-specific fields are also retained. Their interpretation depends on the active valve-area parameterization, so inactive alternative fields should not be mixed into one metering law.

Each command passes through a Simulink–PS Converter configured with unit m. Its stored mode is provide, with user-provided derivative value zero. Although an input-filter time-constant field of 0.005 s exists, this field alone does not demonstrate an enabled first-order spool filter. The controller clips dimensionless-looking numeric commands near unity, whereas the converter declares displacement in metres. The command-to-spool scale therefore needs an explicit unit check against the roughly 5 mm valve travel. Treating the separate legacy custom valve's 0.07 s filter as the active valve dynamics would describe a different implementation.

E. Instrumentation and Data Paths

Pressure taps on both cylinder chambers connect to pressure-sensor subsystems and PS–Simulink converters. Extension sensing comes from the prismatic position signal; force sensing branches from the interface force signal. Arm revolute

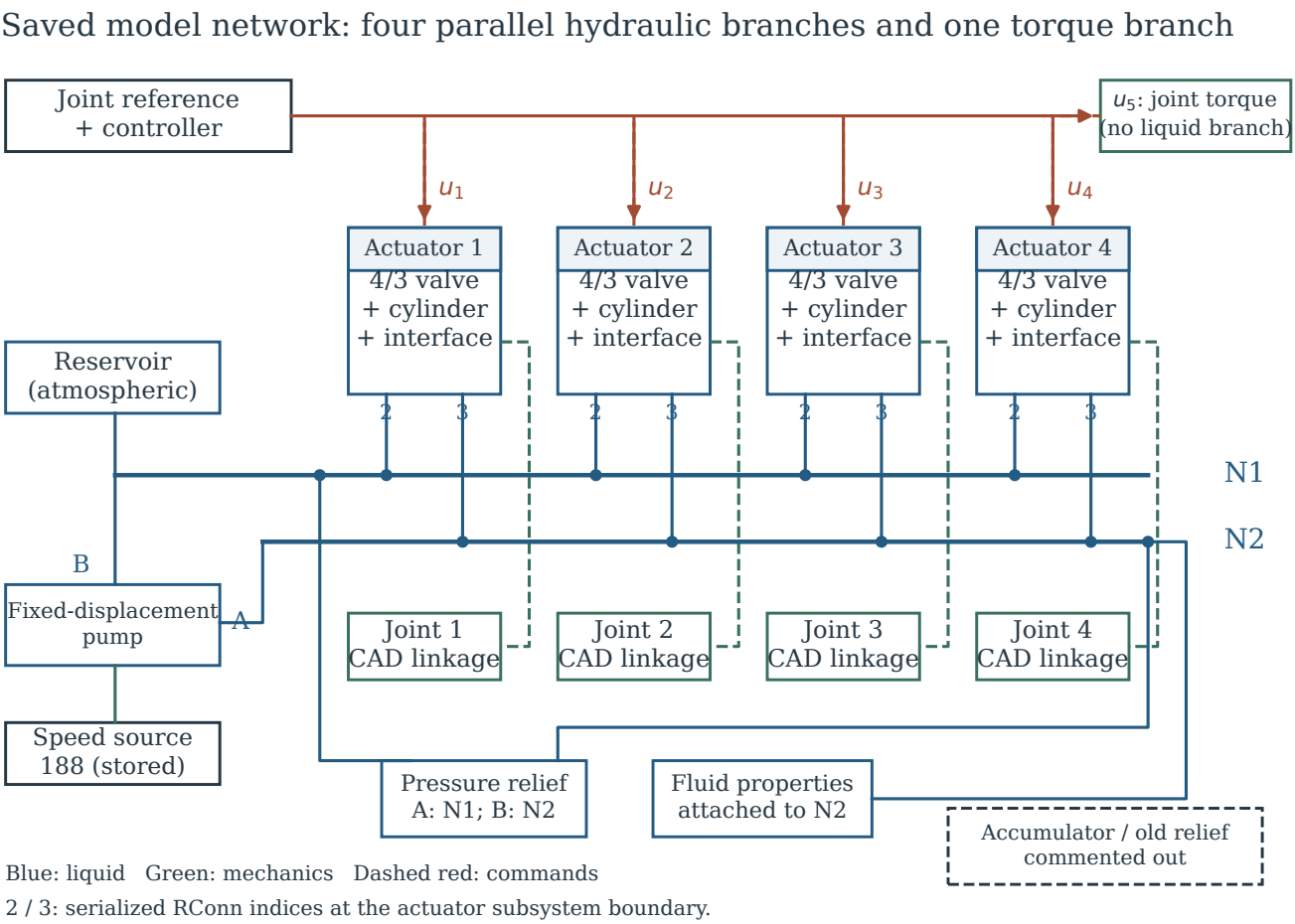


Fig. 6. Complete connection-group reconstruction of the saved model. N1 and N2 name the reservoir-connected and sensor/fluid-connected rails, respectively. Serialized connector indices are retained at subsystem boundaries; they require a compiled port/sign check before being interpreted as a validated operating circuit. Blue lines show liquid connectivity, green denotes mechanical coupling, and dashed red denotes commands. The four hydraulic modules share both rails; the fifth command bypasses the liquid network. Commented components have no active branch. This diagram represents connectivity, not measured flow direction.

joints supply position and velocity. The plant groups these signals into vectors, with stored Unit Delay blocks in the signal-routing subsystem. These discrete delays and the controller's integration interval belong to the measurement/control path, not to the continuous hydraulic chamber equations.

Reference Signal Editor blocks load joint position, velocity, acceleration, and end-effector datasets. The controller uses position and velocity references and measured joint motion; pressure ports exist on the controller function but are unused by its current equations. Supply pressure, chamber pressures, and cylinder strokes are therefore available for diagnostics without constituting pressure feedback control. The root end-effector error functions are commented out in the saved model, so their mere presence does not guarantee an e_norm log in a new run.

F. Assembly and Initialization Procedure

To rebuild the network, first establish one reservoir boundary and the selected fluid-properties and Solver Configuration blocks. Connect the pump shaft, mechanical reference, and speed source; then confirm pump port orientation and the pressure-relief path using pressure and flow signs. Connect one valve/cylinder branch and verify that a small positive and negative spool displacement produce the intended extension and retraction. Only then duplicate the branch for the remaining cylinders and connect the common rails.

Before simulation, reconcile the cylinder's zero-displacement definition with the CAD reference length, evaluate the workspace-dependent offset, and choose compatible chamber pressures and multibody state targets. Enable only the intended friction, leakage, cushioning, and hard-stop contributions. Next, check the command-to-spool conversion, actuator signs, sensor units, and sample times. Close the controller loop after these tests, retain the complete startup transient, and log reference, response, chamber pressures, strokes, valve commands, and tracking error on declared timestamps. This construction procedure explains how to obtain a testable model; it does not substitute for a

TABLE III
DOCUMENTED DESIGN VERSUS ACTIVATION IN THE SAVED EDITABLE MODEL

Component/effect	Saved state	Interpretation
Four valve/cylinder/friction branches	Active replacement blocks	Use these parameters rather than commented predecessors.
Gas-charged accumulator	Commented out	The saved configuration does not provide accumulator buffering.
Older relief and fluid-properties blocks	Commented out	The active relief and Hydraulic Fluid block define the retained configuration.
Built-in cylinder friction/leakage	Disabled on active cylinders	External friction remains active; internal leakage is not enabled there.
Explicit pipe/hose elements	Not present in the main circuit inventory	A separate distributed line-compliance or hose-wave model is not established by this network.
Fifth axis	Direct torque-command conversion	Five commands do not imply five valve-controlled cylinders.
End-effector error functions at root	Commented out	Enable a consistent logging/evaluation path before recomputing tracking errors.

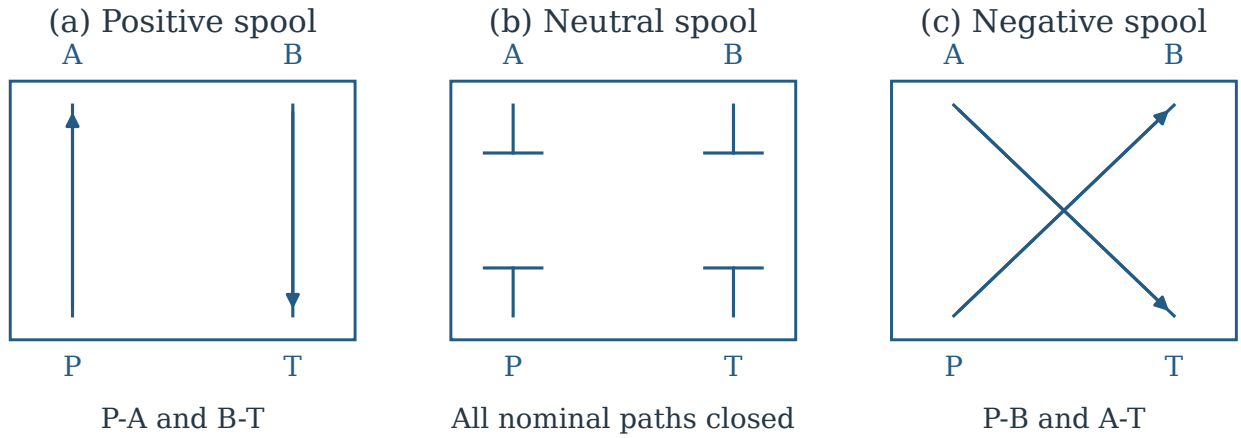


Fig. 7. Metering paths selected by the active 4-way, 3-position valve configuration. Positive spool displacement connects P to A and B to T; negative displacement reverses the chamber connections. At neutral the nominal paths are closed, with a finite leakage-area floor in the block parameterization. Arrows illustrate valve-relative routing under the usual pressure ordering, not a verified flow direction in the saved network.

TABLE IV
SELECTED ACTIVE SUPPLY AND FLUID PARAMETERS

Stored quantity	Value and unit
Pump displacement	25 m^3/rad (saved field)
Pump speed source	188; source unit stored as 1
Relief differential set pressure	27.579 MPa
Relief pressure range	2.759 MPa
Liquid density at atmosphere	845 kg/m^3
Liquid bulk modulus	1.5 GPa
Kinematic viscosity	52.6297 cSt
Stored entrained-air fraction	0.004
Reservoir pressure specification	Atmospheric

successfully executed validation run.

V. HYDRAULIC MODEL

A. Supply and Fluid Compressibility

The active circuit contains a fixed-displacement pump, a reservoir boundary, a relief element, four directional valves, and four double-acting actuators. The accumulator and some earlier alternatives are disabled in the saved model, as summarized in Table III. The governing relations below explain the component behavior; active block options and units determine the particular realization.

For an ideal displacement pump,

$$Q_p = D_p \omega, \tag{12}$$

when displacement D_p is expressed per radian and ω in rad/s [9]. If displacement is specified per revolution, the conversion must be included. An idealized accumulator gas law is $p_g V_g^{\kappa} = \text{const}$ with absolute gas pressure p_g. Its effect depends on precharge, volume, and connection conditions.

Compressibility is described by an effective bulk modulus. A simple mixture approximation for liquid with local gas-volume fraction α is

$$\frac{1}{\beta_{\text{eff}}} \simeq \frac{1-\alpha}{\beta_\ell} + \frac{\alpha}{\kappa p_{\text{abs}}}. \tag{13}$$

All terms have units of inverse pressure. This corrects the dimensionally incomplete volume-weighted expression in the earlier report. The sensitivity study varies a bulk-modulus parameter directly; it does not identify an entrained-air fraction from experiments.

B. Valve Dynamics and Flow

If finite spool dynamics are explicitly enabled, a first-order approximation can be written as

$$\tau_v \dot{x}_v = k_u u - x_v. \tag{14}$$

Here u is the electrical or normalized command and k_u maps it to spool displacement. This conversion is required if u and x_v have different units. This differential equation is a modeling option; it is not asserted to be an enabled spool-lag state in the saved converter configuration. The open flow path follows the signed orifice relation

$$Q = C_d A_o(x_v)\,\text{sgn}(\Delta p)\sqrt{\frac{2|\Delta p|}{\rho}}. \tag{15}$$

The signed form allows pressure reversal. A 4/3 valve selects different supply and return paths for opposite spool directions; its metering area, leakage opening, and limits determine the flow behavior.

The integrated model uses Simscape Fluids isothermal-liquid blocks. Separate custom Simscape prototypes are also supplied. The custom valve uses a 0.07 s first-order command filter and a pressure-root flow law. These prototypes are retained separately because their legacy hydraulic-domain parameters are not automatically identical to the integrated isothermal-liquid implementation.

C. Double-Acting Cylinder

For extension coordinate s, define

$$V_A = V_{A0} + A_A s, \qquad V_B = V_{B0} + A_B(L-s). \tag{16}$$

With Q_A positive into chamber A, Q_B positive out of chamber B, and internal leakage $Q_\ell = C_\ell(p_A - p_B)$, continuity gives

$$\dot{p}_A = \frac{\beta_{\text{eff}}}{V_A}(Q_A - A_A\dot{s} - Q_\ell), \tag{17}$$

$$\dot{p}_B = \frac{\beta_{\text{eff}}}{V_B}(-Q_B + A_B\dot{s} + Q_\ell). \tag{18}$$

The pressure-force relation is

$$F_h = A_A p_A - A_B p_B - F_f(\dot{s}). \tag{19}$$

For this force relation, pressures are relative to the cylinder environment. With absolute pressures, subtract $(A_A - A_B)p_{\text{env}}$ from the pressure-force term; this correction cancels for the saved equal-area blocks. Area asymmetry, dead volume, and piston motion affect the chamber pressures [6].

One regularized viscous/Coulomb friction model is

$$F_f(v) = b_v v + F_c \tanh(v/v_\epsilon). \tag{20}$$

The archived custom cylinder implements this regularized form with an optional efficiency multiplier and no explicit internal-leakage term. It is separate from the active stock actuator plus external Translational Friction blocks. The latter provide their own breakaway, Coulomb, and viscous parameterization; the active actuator's internal friction/leakage switches are off. The general equations above describe the physical framework, while Table II and the enabled block options identify the saved realization.

VI. Trajectory and Feedback Control

A. Reference Generation

The reference pipeline uses a rigid-body representation and inverse kinematics to generate joint positions, velocities, accelerations, and an end-effector path. The project includes an arc-and-chord trajectory generator as well as a separate mowing-sweep example. These are different reference-generation examples and should not be treated as one experiment.

The integrated model's Signal Editor blocks read the supplied joint-position, joint-velocity, joint-acceleration, and end-effector MAT datasets. The organized project retains these files without altering their numerical content. The stored input sample times are 0.005 s for position, velocity, and end-effector references and 0.01 s for acceleration. Dataset timing and controller execution must be checked together before a new run.

B. Archived Controller Structure

The original report describes integral backstepping on every axis. The embedded controller in the archived integrated model is more specific: joints 1, 3, 4, and 5 use a backstepping-inspired error law, while joint 2 uses position PI, velocity feedback, gravity bias, and a soft limit term. Pressure and supply-pressure ports are present but unused by that function.

For the four backstepping branches,

$$z_{1,i} = q_{r,i} - q_i, \tag{21}$$

$$\alpha_i = \dot{q}_{r,i} + c_{1,i} z_{1,i}, \tag{22}$$

$$z_{2,i} = \alpha_i - \dot{q}_i, \tag{23}$$

$$I_i^+ = \text{clip}(I_i + T_s z_{2,i}, -I_{\max,i}, I_{\max,i}), \tag{24}$$

$$u_{b,i} = \frac{z_{1,i} + c_{2,i} z_{2,i} + K_{I,i} I_i}{b_i}. \tag{25}$$

Integration is gated when the previous command approaches saturation. A friction-bias term is added above a small error threshold; actuator sign and command bias are then applied before clipping. This practical error-feedback law is not accompanied by a Lyapunov proof for the full coupled plant.

Joint 2 instead uses

$$u_{\text{PI},2} = K_p e_2 + K_i I_2 + K_v(\dot{q}_{r,2} - \dot{q}_2), \tag{26}$$

TABLE V
EMBEDDED CONTROLLER SETTINGS FOR BACKSTEPPING BRANCHES

Joint	c_1	c_2	K_I	b	$\lvert u\rvert_{\max}$
1	800	1	1.5	8	0.95
3	6	1.5	2	12	0.80
4	800	1	2	10	0.80
5	6	1.5	1	15	0.90

with $K_p = 1200$, $K_i = 3$, and $K_v = 4$. A configuration-dependent bias $0.40\cos q_2 + 0.25\cos(q_2 + q_3)$ and a soft position-limit contribution are added before actuator mapping and saturation. These are command-space gains and biases; they are not identified physical torque parameters.

The embedded integration interval is $T_s = 0.005$ s, rather than the 0.01 s stated in the earlier report. The model's separate error-accumulation helper contains 0.01 s. The offline metric function uses the actual recorded time vector and trapezoidal integration, which avoids silently assigning the controller's interval to every log. A controller time constant stored in code is not proof that the MATLAB Function executes at that interval; reference sample times, inherited rates, and routing delays must be checked together.

VII. Evaluation Method

A. Threshold-Based Response Timing

The MATLAB analysis function monitors four channels using valve command, cylinder extension, chamber pressures, and joint position. Command onset t_0 is the first threshold crossing of $|u(t) - u(0)|$, with threshold equal to the larger of 5% of the command span and an absolute value of 0.002 in the command's units. This is distinct from the 50% criterion used for the response signals.

For a response r, the analysis finds the first post-onset crossing of

$$|r(t) - r(0)| = 0.5 \max_{t \geq t_0} |r(t) - r(0)|. \tag{27}$$

Cylinder extension, differential pressure, and joint position define times t_x, t_p, and t_q. The principal intervals are

$$\tau_{v\to x} = t_x - t_0, \quad \tau_{v\to p} = t_p - t_0, \quad \tau_{v\to q} = t_q - t_0. \tag{28}$$

Crossings are linearly interpolated between recorded samples. This improves the numerical crossing estimate without creating independent sub-sample measurement resolution. If a response is already above threshold at the first retained post-onset sample, the implementation uses that sample's time.

The quantity $\tau_{v\to q}$ is a threshold-based rise-response interval. It combines command dynamics, pressure response, reference shape, controller action, and mechanical motion; it is not a separately identified pure transport delay. Nonmonotone trajectories and small-amplitude axes require care in interpreting their first crossing.

For four available joint intervals, the reported aggregate is

$$\tau_{\mathrm{RMS}} = \sqrt{\frac{1}{4}\sum_{j=1}^{4} \tau_{v\to q,j}^2}. \tag{29}$$

An undefined or unexcited response should be marked missing and reported with its valid count. It must not be silently replaced by a zero-delay measurement.

B. Task-Space Error

For synchronized reference and response positions,

$$e(t) = \|p_r(t) - p(t)\|_2. \tag{30}$$

On a declared active window of duration T,

$$E_{\mathrm{RMS}} = \sqrt{\frac{1}{T}\int_{t_s}^{t_e} e(t)^2\, dt}, \tag{31}$$

$$E_{\mathrm{mean}} = \frac{1}{T}\int_{t_s}^{t_e} e(t)\, dt. \tag{32}$$

The offline function begins at the earliest valid command onset and uses time-weighted trapezoidal integration. A variable-step solver makes time weighting particularly relevant. The archived spatial curves in Fig. 8 have different sample counts and lack a shared time base; indexwise subtraction of these extracted curves would therefore not reproduce these metrics.

VIII. Results and Discussion

A. Available Numerical Evidence

The four reported cases use $\beta \in \{0.5, 1.0, 1.5, 2.0\}$ GPa. The archive does not bind each case to a complete evaluated block-parameter record, and the saved active replacements differ from some earlier presentation settings. The reported cases therefore remain historical summaries rather than verified reruns of the present saved configuration. Table VI and Fig. 9 reproduce the supplied numerical summaries. The values agree with the presentation, Table I of the earlier report, and delaysvserror.m; a conflicting joint-3 series in the earlier Table II is documented separately.

B. Joint-Specific Trends

Joint 1 has the largest reported response interval. Raising the bulk modulus from 0.5 to 2.0 GPa reduces that interval by 197.7 ms, or approximately 23.3%. The reported aggregate decreases by 93.8 ms, approximately 21.1%. Most of the reduction occurs below 1.5 GPa in these four cases.

The other joints do not follow an identical trend. Joint 3 decreases from 186.3 to 149.9 ms at 1.5 GPa, then rises to 152.4 ms. Joint 4 increases from 202.1 to 221.6 ms, about 9.6%. This nonuniform behavior is consistent with the fact that a threshold interval depends on the coupled plant, feedback law, and trajectory. The results do not support claiming that every axis becomes faster as fluid stiffness increases.

The earlier report states that joint 2's interval remains below 0.3 ms. Its four individual values are not included in the summary script. The aggregate can be approximated to the displayed precision by setting this small contribution to zero; however, that arithmetic is not evidence of a truly instantaneous joint response. It also cannot establish a valve bandwidth from signals sampled on a coarser grid.

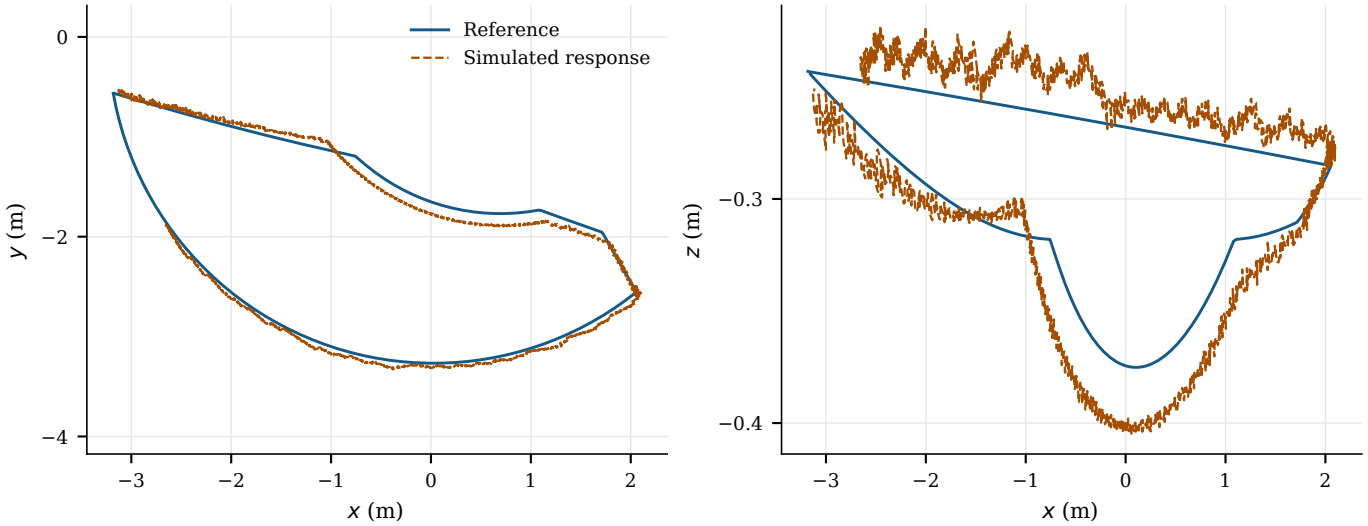


Fig. 8. Archived task-space reference and simulated response, redrawn directly from numerical curves in plot.fig. Left: horizontal projection; right: vertical projection. The 3,991 reference and 3,550 response samples lack a shared time vector; these spatial traces cannot independently reproduce a tracking-error metric.

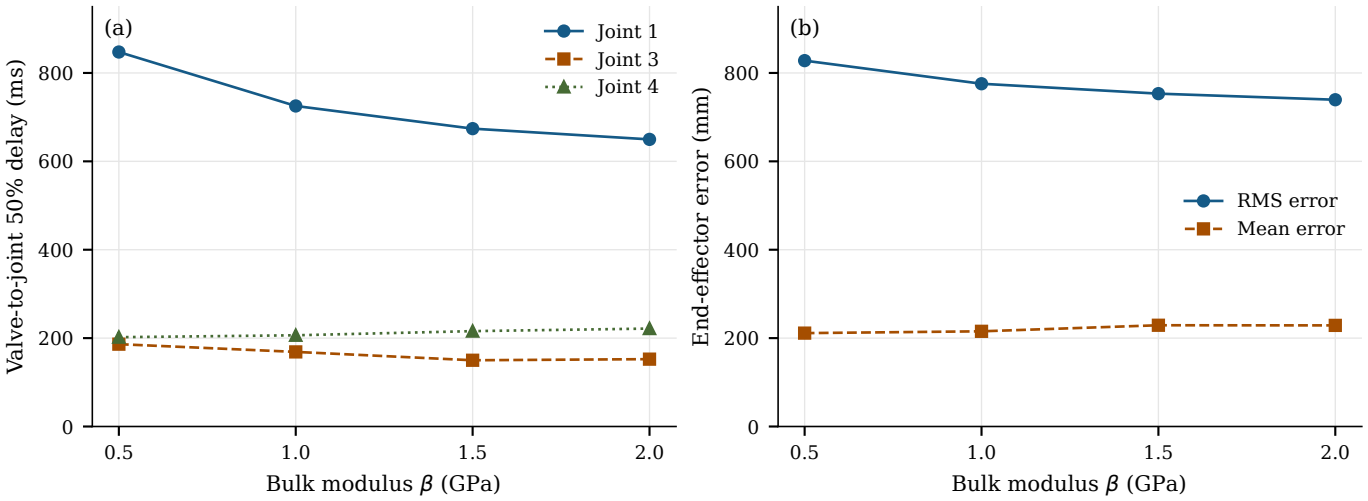


Fig. 9. Reported bulk-modulus sensitivity redrawn from the numerical values in the original analysis script and report. (a) Joint-specific 50% response intervals. (b) RMS and mean end-effector errors. Markers denote the four reported cases; connecting lines guide the eye and are not extra simulated cases.

C. Tracking-Error Trends

The reported RMS end-effector error falls from 827.9 to 739.4 mm, a reduction of 88.5 mm or approximately 10.7%. The mean error does not improve in parallel: it changes from 211.3 to 228.8 mm, approximately 8.3% higher at the largest modulus. The earlier statement that the mean error follows the same improving pattern is therefore incorrect.

RMS and mean error emphasize different parts of a trajectory. A lower RMS with a higher mean is mathematically possible, for example when large transient errors decrease while a smaller persistent error increases. The available summary values alone do not establish which time segments cause the difference. Their relatively large RMS-to-mean ratios motivate inspecting synchronized error histories, initialization, and the active integration window in future runs.

The largest reported RMS error is about 0.83 m and the smallest about 0.74 m. These values should be presented with their physical scale; they do not justify an unqualified claim of high-precision mowing. The spatial view in Fig. 8 helps illustrate the trajectory, but it cannot replace quantitative error analysis.

D. Scope of Validation

The sweep establishes sensitivity of the simulated closed-loop system to a selected parameter. It does not independently verify that chamber pressures, friction, cylinder forces, and end-effector motion match a physical mower. The archive supplies no synchronized hardware measurements or parameter-identification uncertainty bounds for such a comparison.

TABLE VI
REPORTED SIMULATION SUMMARY ACROSS BULK-MODULUS CASES

β (GPa)	Joint 1 (ms)	Joint 3 (ms)	Joint 4 (ms)	Reported τ_{RMS} (ms)	E_{RMS} (mm)	E_{mean} (mm)
0.5	847.4	186.3	202.1	445.4	827.9	211.3
1.0	725.3	168.8	206.4	386.4	775.6	215.4
1.5	674.0	149.9	215.8	361.7	753.2	229.1
2.0	649.7	152.4	221.6	351.6	739.4	228.8

Similarly, the smooth normalized step responses in the original delay_plots.m script are generated from an assumed second-order model, fitted to reported crossing times. They are explanatory reconstructions, not raw hydraulic-simulation traces. They remain in the archive but are excluded from the revised report's result figures. The separate custom components and parameter helpers describe additional configurations; their constants should not be mixed with the integrated model's blocks as if all came from one run.

IX. REPRODUCIBILITY AND FURTHER WORK

The repository retains the editable R2023b model, canonical meshes, reference datasets, analysis scripts, and numeric plot exports. Project-relative paths resolve geometry and Signal Editor files, while generated products go into a separate build directory. Equations, controller code, and solver settings are unchanged.

A machine-readable specification records selected block settings, units, activation flags, XML members, and connection groups. Structural checks verify retained hashes, model archives, paths, numeric curves, and documentation. These checks do not compile or run MATLAB. No end-to-end simulation or hardware test was conducted for this report revision.

Further work should first reconcile the pump displacement unit, supply/return port associations, command-to-spool scale, effective cylinder areas, large friction coefficients, initialization offsets, and final-joint travel constraint. These are configuration checks, not a diagnosis that any one item explains a particular recorded motion. Then establish one versioned model configuration and timing contract. A reproducible sweep should save the controller gains, active block parameters, solver settings, reference trajectory, all event signals, and synchronized end-effector errors for each case. Tolerance studies should demonstrate convergence of the reported metrics. Physical validation should then compare measured and simulated pressure, stroke, force, and motion under matched inputs before the model is described as a calibrated digital twin.

X. CONCLUSION

The project constructs a coupled arm from CAD bodies, alignment joints, force-actuated prismatic joints, and a shared four-branch hydraulic network. A fifth command acts directly on the final revolute joint. The detailed actuator construction explains how chamber pressure generates axial force and how multibody position/velocity feed back into the hydraulic states. Geometry-dependent force transmission and fluid continuity connect the two physical domains. The reported parameter sweep shows that larger bulk modulus reduces the dominant joint's response interval and end-effector RMS error for the supplied task, while other joint intervals and mean error behave differently. These results motivate evaluating several metrics together and retaining the complete experimental configuration. The organized model and numerical assets provide a documented basis for further simulation, identification, and control development; agreement with physical hardware remains a separate validation task.

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